In particular, $-iI_d \in M_d(\mathbb{C})$ corresponds to the block matrix $J_{2d_1} \oplus J_{2d_2}$. Note that, under both identifications, the adjoint of a complex matrix corresponds to the transpose of its identification.

2.1. **Symplectic and Orthosymplectic Matrices.** By a *symplectic transformation* of $\mathbb{C}^d$ we mean a real linear operator L on $\mathbb{C}^d$ that preserves the standard symplectic form, i.e.,

$$\operatorname{Im}\langle L\mathbf{u}|L\mathbf{v}\rangle = \operatorname{Im}\langle \mathbf{u}|\mathbf{v}\rangle \qquad \forall\ \mathbf{u},\mathbf{v} \in \mathbb{C}^d. \tag{1}$$

Now, fix an identification of $\mathbb{C}^d$ with $\mathbb{R}^{2d}$, as described in the previous section. This identification allows us to represent the real linear transformation L as a $2d \times 2d$ real matrix (again denoted as L). Consider a specific $J \in \{J_{2d}, J_{2d_1} \oplus J_{2d_2}\}$, where $d = d_1 + d_2$. It is important to note that $\mathbf{z}^T J\mathbf{z}' = \operatorname{Im}\langle \mathbf{z}|\mathbf{z}'\rangle$ for all $\mathbf{z},\mathbf{z}' \in \mathbb{C}^d = \mathbb{R}^{2d}$. Therefore, the equation (1) is equivalent to stating that $L^T JL = J$. The elements of the set

$$Sp(2d, J) := \{L \in M_{2d}(\mathbb{R}) : L^T JL = J\}$$

are called J-*symplectic* matrices. A J-symplectic matrix that is also orthogonal is called a J-*orthosymplectic* matrix. The set $Sp(2d, J)$ forms a group under multiplication, called J-*symplectic group*. Clearly, $J \in Sp(2d, J)$. Every J-symplectic matrix L has determinant 1, and its inverse is given by $L^{-1} = J^T L^T J$. Note also that $L \in Sp(2d, J)$ implies $L^T \in Sp(2d, J)$. A J_{2d}-symplectic matrix and a J_{2d}-orthosymplectic matrix are simply called a *symplectic* matrix and an *orthosymplectic* matrix, respectively. Note that if $L = \begin{bmatrix}\mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_{2d}\end{bmatrix} \in M_{2d}(\mathbb{R})$ and $J = [J_{ik}] \in M_{2d}(\mathbb{R})$, then

$$L \in Sp(2d, J) \iff \mathbf{u}_i^T J\mathbf{u}_k = J_{ik}, \qquad \forall\ 1 \le i,k \le 2d. \tag{2}$$

Consequently, a matrix $L = \begin{bmatrix}\mathbf{u}_1, \cdots, \mathbf{u}_d, \mathbf{v}_1, \cdots, \mathbf{v}_d\end{bmatrix} \in M_{2d}(\mathbb{R})$ is a symplectic if and only if its columns form a symplectic basis.

2.2. **Weyl Representation and Bogoliubov Transformations.** The *symmetric Fock space* over $\mathbb{C}^d$ is defined by

$$\Gamma(\mathbb{C}^d) := \bigoplus_{k=0}^{\infty} (\mathbb{C}^d)^{\circledS^k},$$

where $(\mathbb{C}^d)^{\circledS^k}$ is the k-times *symmetric tensor product* of $\mathbb{C}^d$ (see [Par92, Chapter 2] for details). For $\mathbf{v} \in \mathbb{C}^d$, the *exponential vector* $|e(\mathbf{v})\rangle$ is defined as

$$|e(\mathbf{v})\rangle := \bigoplus_{k=0}^{\infty} \frac{\mathbf{v}^{\otimes^k}}{\sqrt{k!}},$$

where $\mathbf{v}^{\otimes^0} := 1 \in \mathbb{C}$. The span of exponential vectors forms a dense subset of $\Gamma(\mathbb{C}^d)$. For $\mathbf{u} \in \mathbb{C}^d$, the *Weyl unitary operator*, $W(\mathbf{u})$ is the unique unitary operator on $\Gamma(\mathbb{C}^d)$ satisfying

$$W(\mathbf{u})\,|e(\mathbf{v})\rangle = \exp\left\{-\frac{1}{2}\|\mathbf{u}\|^2 - \langle \mathbf{u}|\mathbf{v}\rangle\right\}|e(\mathbf{u}+\mathbf{v})\rangle, \qquad \forall\ \mathbf{v} \in \mathbb{C}^d.$$

The adjoint of $W(\mathbf{u})$ is given by $W(\mathbf{u})^\dagger = W(-\mathbf{u})$. The Weyl unitary operators satisfy the Weyl form of the the *canonical commutation relations*:

$$W(\mathbf{u})W(\mathbf{v}) = \exp\{-i\operatorname{Im}\langle \mathbf{u}|\mathbf{v}\rangle\}W(\mathbf{u}+\mathbf{v}), \qquad \forall\ \mathbf{u},\mathbf{v} \in \mathbb{C}^d;$$
$$W(\mathbf{u})W(\mathbf{v}) = \exp\{-2i\operatorname{Im}\langle \mathbf{u}|\mathbf{v}\rangle\}W(\mathbf{v})W(\mathbf{u}), \qquad \forall\ \mathbf{u},\mathbf{v} \in \mathbb{C}^d.$$